

Two-Step Equations in Context

MA.7.AR.2.2

I can solve a two-step equation from a situation, check it, and say what the answer means.

Same equations as yesterday.

Today they are about something, and the answer is a sentence.

Warm-Up

On your own. Four minutes.

Same context, different question

A gym charges a \$45 joining fee and then \$18 a month.

- 1.** What does it cost for 6 months?
- 2.** After how many months has Dario paid \$243?
- 3.** What does the 45 stand for, and what does the 18 stand for?

Notes

I. Say what the letter means before you solve

- A. **Write down what the letter stands for, in words, first.**
- B. The thing that CHANGES is the variable. The thing that stays the same is the constant.
- C. **After you solve, answer in a sentence with units.**

Notes

II. Solve, check, interpret — every time

- A. **Solve.** Undo the addition or subtraction, then the multiplication or division.
- B. **Check.** Substitute into the original equation.
- C. **Interpret.** Write a sentence saying what the number means.

Notes

III. Dividing first works — if you divide everything

Wilma and Maurio both solved $42 = -3x + 15.6$.

Wilma: subtract first	Maurio: divide first
$42 - 15.6 = -3x$	$42 \div (-3) = x + 15.6 \div (-3)$
$26.4 = -3x$	$-14 = x - 5.2$
$x = -8.8$	$x = -8.8$

Notes

IV. A negative answer is information, not a mistake

- A. Sometimes the answer comes out negative and the situation allows it — a temperature, a debt, a depth.
- B. **Sometimes it does not. A time, a length or a count cannot be negative.**
- C. **When it cannot be, the equation was set up wrong. Go back to the setup, not the arithmetic.**

Example 1

Dan is fencing three sides of his backyard.

He has 80 feet of fencing. The yard is 48.6 feet long.

How wide can the fenced part be?

What does your letter stand for? Say it before you write anything.

Example 1

Solve and check.

$$48.6 + 2w = 80$$

$$2w = 31.4$$

$$w = 15.7$$

$$\text{Check: } 48.6 + 2(15.7) = 48.6 + 31.4 = 80 \quad \checkmark$$

Example 1

Now say what it means.

The fenced part can be 15.7 feet wide.

Example 2

A runner's plan.

She warms up for 15 minutes, then runs at 6 minutes a mile.

Her whole session is 45 minutes. How many miles does she run?

Example 2

Solve it.

$$15 + 6m = 45$$

$$6m = 30$$

$$m = 5$$

$$\text{Check: } 15 + 6(5) = 15 + 30 = 45 \quad \checkmark$$

She runs 5 miles.

Example 2

And if the total had been 12 minutes?

$$15 + 6m = 12$$

$$6m = -3$$

$$m = -0.5$$

Example 3

Three students. Each one knew the method.

The step as written	What went wrong	Correct
$-6t = 36$, then divide by 6		
$20 = 4y$, then subtract 4		
$-2(-5) = -2(-\frac{1}{2}x) - 3$		

Example 3

What went wrong.

The step as written	What went wrong	Correct
$-6t = 36$, then divide by 6	Divided by 6 instead of -6	$t = -6$
$20 = 4y$, then subtract 4	Subtracted instead of dividing	$y = 5$
$-2(-5) = -2(-\frac{1}{2}x) - 3$	Did not multiply every term by -2	$x = 4$

Example 3

One term each.

Every student here knew the method. Each lost it on one term.

- A. A legal property applied to only part of an expression is not a legal move.
- B. **Row 3 is the whole of Lesson 5: distributing means every term inside, every time.**

Whiteboards • Question 1 of 9

Boards up on three.

$$15 + 6m = 45$$

Solve it, and say what the answer means.

A number and a sentence.

Whiteboards • Question 1 of 9

Answer.

$$15 + 6m = 45$$

$$6m = 30$$

Answer: $m = 5$

Whiteboards • Question 2 of 9

Boards up on three.

$$15 + 6m = 46$$

Solve it, and say what the answer means.

A number and a sentence.

Whiteboards • Question 2 of 9

Answer.

$$15 + 6m = 46$$

$$6m = 31$$

$$\textbf{Answer: } m = 31/6 \approx 5.17$$

Whiteboards • Question 3 of 9

Boards up on three.

$$15 - 6m = 45$$

Solve it, and say what the answer means.

A number and a sentence.

Whiteboards • Question 3 of 9

Answer.

$$15 - 6m = 45$$

$$-6m = 30$$

$$\textbf{Answer: } m = -5$$

Whiteboards • Question 4 of 9

Boards up on three.

$$15 - 6t = 45, \quad t = \text{degrees change}$$

Solve it, and say what the answer means.

A number and a sentence.

Whiteboards • Question 4 of 9

Answer.

$$15 - 6t = 45, \quad t = \text{degrees change}$$

same arithmetic, different meaning

Answer: $t = -5^\circ$, and that is fine

Whiteboards • Question 5 of 9

Boards up on three.

$$6m + 15 = 45$$

Solve it, and say what the answer means.

A number and a sentence.

Whiteboards • Question 5 of 9

Answer.

$$6m + 15 = 45$$

the terms are swapped; nothing else is

Answer: $m = 5$

Whiteboards • Question 6 of 9

Boards up on three.

$$6(m + 15) = 45$$

Solve it, and say what the answer means.

A number and a sentence.

Whiteboards • Question 6 of 9

Answer.

$$6(m + 15) = 45$$

divide by 6 first: $m + 15 = 7.5$

Answer: $m = -7.5$

Whiteboards • Question 7 of 9

Boards up on three.

Renting a kayak costs \$24 for the first 3 hours and \$4.50 for every hour after that.
You spent \$42. $24 + 4.50m = 42$

Solve it, then say how long you had the kayak.

A number and a sentence.

Whiteboards • Question 7 of 9

Answer.

Renting a kayak costs \$24 for the first 3 hours and \$4.50 for every hour after that.

You spent \$42. $24 + 4.50m = 42$

$$4.50m = 18, \text{ so } m = 4$$

Answer: 4 extra hours — 7 hours in total

Whiteboards • Question 8 of 9

Boards up on three.

A gym charges \$55 to join plus a monthly fee. A full year costs \$235. $55 + 12t = 235$

Solve it, then say what t means.

A number and a sentence.

Whiteboards • Question 8 of 9

Answer.

A gym charges \$55 to join plus a monthly fee. A full year costs \$235. $55 + 12t = 235$

$$12t = 180, \text{ so } t = 15$$

Answer: \$15 — the monthly fee

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Alishia solved $7x - 4\frac{1}{4} = 44.75$ and got $x = 5.75$. Check it, then solve it correctly.

Write your answer in sentences.

This one is written work. Show the substitution.

Whiteboards • Question 9 of 9

Answer.

Alishia solved $7x - 4\frac{1}{4} = 44.75$ and got $x = 5.75$. Check it, then solve it correctly.

$$7(5.75) - 4.25 = 36, \text{ not } 44.75$$

Answer: $x = 7$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.